

Kimberly Wang

San Jose, CA / Troy, NY |  | [linkedin.com/in/wangkimb/](https://www.linkedin.com/in/wangkimb/) | kimbwang.github.io

EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. Electrical Engineering, GPA: 3.7

Aug 2024 - Expected Dec 2027

Coursework: Electric Circuits, Computer Components and Operations, Embedded Control, Intro to Electronics, Intro to Engineering Design, Signals and Systems

TECHNICAL SKILLS

Hardware: Arduino, MSP432, ESP32, Basys 3, Oscilloscope, Waveform Generator, Spectrum Analyzer, Multimeter

Software: LTspice, Cadence Allegro, Altium Designer, Xilinx Vivado, MATLAB, LabVIEW, Simulink, Office Suite

Languages: Python, C/C++, SQL, VHDL, MATLAB

EXPERIENCE

Hardware Developer Intern

Starting May 2026

IBM

Poughkeepsie, NY

- Incoming hardware developer intern working on the IBM Z Mainframe I/O Hardware

Undergraduate Student Researcher @ Mercer XLab

Jan 2025 - Present

Rensselaer Polytechnic Institute

Troy, NY

Software Configured Breadboard (SCB) Supervised by ADI Fellow Douglas Mercer and Prof. Shayla Sawyer

- Used Python and ALICE software to computerize the wiring of breadboards to reduce debugging and wiring issues
- Executed board-level validation and root-cause analysis utilizing automated Python scripts and a multimeter
- Authored technical documentation and user guides for functions and processes
- Collaborating with RPai to utilize Cursor AI to automate circuit design, building, and testing

Undergraduate Student Assistant

Aug 2025 - Present

Rensselaer Polytechnic Institute

Troy, NY

- “Intro to ECSE” (F25), “Computer Components and Operations” (SP26)
- Advise on analog and digital circuit building, testing, and measurement
- Clarify questions about theoretical concepts within contexts of labs

PROJECTS

CO Sensor System | *LTspice, Benchtop Equipment*

- Created a multi-stage analog system for maintaining safety from CO in buildings
- Designed and tested complete analog circuit for indication, ventilating, and alarming systems
- Performed failure analysis and problem solving to ensure functionality of physical circuit

Compass Controlled TI RSLK Car | *MSP432, C, I2C*

- Designed a closed-loop navigation system using a magnetometer via I2C protocols for real time directions
- Implemented PID controls in C to minimize errors in motor movements
- Configured GPIO interrupts to handle data acquisition and maintain stable motor control

Implementation of ALU to Demonstrate Boolean Functions | *Vivado, VHDL, Basys 3 (FPGA)*

- Authored VHDL code to write logic for an ALU with 4 bits for an adder, subtractor, bit parity, and comparator
- Used Vivado to define constraints and generate bitstream to be sent to Basys 3 FPGA board
- Tested functions with on-board switches as inputs and LEDs for outputs

Puzzle Solving Autonomous Raspberry Pi-Based Cart | *Python, Raspberry Pi*

- Designed a strategy to determine a path based on input on adjacent grids to reach an unknown target grid coordinate and travel back to the starting position
- Program the cart in Python to travel within a physical 4x4 grid with sensor inputs and motor control

PI Controller Design | *LTspice, Oscilloscope*

- Designed and simulated an analog PI controller in LTspice to achieve a steady state voltage in 4s
- Verified the design by building and testing the circuit to get results with < 5% error from simulation